



Technical Service Bulletin

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CES14648 – Coolant Specification Introduction

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

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Issue

Organic Additive Technology (OAT) Coolants have a broad range of base and secondary formulas that meet Cummins Engineering Standard (CES) CES14603. Some formulas that meet CES14603 have been shown to **not** be capable of protection against liner cavitation in Cummins Inc High Horsepower (HHP)

engines with Top Stop Wet Cylinder Liners. The incompatibility can be caused by two main factors:

- Omission of critical chemical additives that are required for cavitation protection
- Inclusion of chemicals that are **not** fully compatible with Cylinder Liner materials.

Verification

CES14648 has been developed around HHP Top Stop Wet Liners to improve cylinder liner protection against cavitation. This CES has been applied to Industrial and Marine applications **only**. A full list of affected engine platforms is available in the following manuals and procedures.

- 5411406 –Fluids for Cummins® Products Service Manual > Section 5 – Coolant:
 - (00-379-004) Engine Coolant Requirements
- All Cummins Inc. Engine Platform Owners Manuals > Section V – Maintenance Specifications:
 - (00-018-004) Coolant Recommendations and Specifications

Reason for Change

Cummins has conducted substantial rig and field testing of Nitrite-Free propylene Glycol based coolants. The results of this testing have determined:

- Propylene glycol based coolants that contain nitrite or nitrite and molybdate additives within a certain concentration range have been shown to be effective in preventing liner cavitation in HHP Engines with Top Stop Wet Cylinder Liners.
- Propylene glycol based coolants that contain nitrite or nitrite and molybdate additives below specific concentration limits, or that omit these additives entirely, have been found to lead to premature liner cavitation in HHP Engines with Top Stop Wet Liners.

As a result of this testing, Cummins has introduced a new Cummins Engineering Standard (CES) CES14648 for coolants used in HHP Engines with Top Stop Wet Liners. CES 14648 requires propylene glycol based coolants to contain nitrite or nitrite and molybdate at the concentrations specified by ASTM D6210, Table A1.1.

CES 14648 permits the use of ethylene glycol based coolants that do **not** contain any nitrite or molybdate.

Visit the CES 14648 registered products page on QSOL to select an appropriate coolant.

Note :

https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/coolant_registration.html
(https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/coolant_registration.html)

Customer Communication

CEAF 202605200001 submitted and issued.

Fluids for Cummins Engines Manual, Section 5, (00-379-004) Engine Coolant Requirements and (00-379-006) Maintenance of Coolant and Cooling System. All Cummins Inc Engine Platform Owners Manuals, Section V – Maintenance Specifications, (00-018-004) Coolant Recommendations and Specifications are updated.

- These manuals serve as "high level" documents. The lower-level service manual documents (Owner, Operation and Maintenance, Service) in all individual engine configurations contain conflicts with the updated high-level documents. Where these conflicts occur, the High-Level documentation takes precedence.

Service Instructions

Follow Service Manual guidelines. Propylene glycol (PG) based coolants **must** contain nitrite or nitrite and molybdate within the specified concentration limits in ASTM D6210 Table A1.1. PG based OAT coolants **not** meeting these requirements should be removed at the earliest opportunity and replaced with a compliant coolant.

Fluids for Cummins Engines Manual, Section 5, (00-379-004) Engine Coolant Requirements and (00-379-006) Maintenance of Coolant and Cooling System. All Cummins Inc Engine Platform Owners Manuals, Section V – Maintenance Specifications, (00-018-004) Coolant Recommendations and Specifications are updated as these serve as 'high level' documents. The lower-level service manual documents (Owner, Operation and Maintenance, Service) in all individual engine configurations contain conflicts with the updated high-level documents. Where these conflicts occur, the High-Level documentation takes precedence.

Publications Affected

Table 1, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Fluids for Cummins Engines Manual	See Affected Product List	5411406	Engine Coolant Requirements	00-379-004	5
Fluids for Cummins Engines Manual	See Affected Product List	5411406	Maintenance of Coolant and Cooling System	00-379-006	5
Owners Manuals	All Cummins Inc Owners Manuals within Technical Service Bulletin, Cummins® Service Model Identification, TSB130080		Maintenance Specifications	00-018-004	V

Document History

Date	Details
2026-5-20	Module Created
2026-5-26	Updated Service Instructions section.
2026-5-27	Updated some wording throughout TSB for clarity.
2026-7-7	Added QSK78 CM500 to TSB.
2026-7-20	Added GTA855 CM558 G101 to product affected list.

